

MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

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Outline

- 1 Real Analysis Lecture 10
 - Compactness
 - Metric Spaces

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Warm-up Challenge

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- Lindelöf Covering Theorem

Bolzano-Weierstrass Theorem

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Theorem (Bolzano-Weierstrass Theorem)

A bounded set $A \subseteq \mathbb{R}^n$ with infinitely many points must have an accumulation point.

Cantor Intersection Theorem

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Suppose that $C_1, C_2, C_3, \dots$ are non-empty closed, bounded sets with $C_{i+1} \subseteq C_i$ for all $i \in I = \mathbb{Z}_+$. Then $\bigcap_{i \in I} C_i$ is also non-empty.

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Therefore the Bolzano-Weierstrass Theorem tells us it has an accumulation point $\vec{x}$.



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Moreover, $A_m \subseteq C_m$ so $\vec{x}$ is an accumulation point of C_m for all m .

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Since C_m is closed, it follows $\vec{x} \in C_m$, and thus $\vec{x} \in \bigcap_{i \in I} C_i$. In particular the intersection is non-empty! □

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$$C_m = A \cap \left(\mathbb{R}^n \setminus \bigcup_{i=1}^m U_i \right).$$

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Since $\vec{x} \in A \subseteq \bigcup_{i \in I} U_i$, we have that $\vec{x} \in U_k$ for some $k \in I$, meaning $\vec{x} \notin C_k$.

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This is a contradiction.



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Metric space

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The value $d(x, y)$ is called a **metric** and describes the “distance” between x and y .

Open balls

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Definition

Let (M, d) be a metric space. The open ball of radius $r > 0$ centered at $x \in M$ is

$$B_M(x; r) = \{y \in M : d(x, y) < r\}.$$

Examples of metrics on $\mathbb{R}^2$

$$d_{\text{eucl}}(\vec{x}, \vec{y}) = |\vec{x} - \vec{y}| = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

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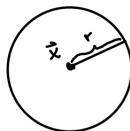
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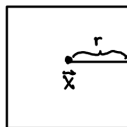
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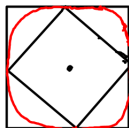
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Challenge!

Let M be a nonempty set and define $d : M \times M \rightarrow \mathbb{R}$ by

$$d_{\text{disc}}(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

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Prove that d_{disc} is a metric.

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Prove that d_{disc} is a metric.

This is called the **discrete metric**.

Challenge!

Problem

What do the open balls with the discrete metric look like?

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The set $\text{int}(A)$ of interior points of A is called the **interior** of A .

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The set A is **open** if every point in A is an interior point, or equivalently $\text{int}(A) = A$.

Challenge!

Problem

Consider the metric space $(\mathbb{R}, d_{\text{disc}})$ where d_{disc} is the discrete metric.

What sets are open sets?

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A point $x \in M$ is an **adherent point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A .

A point $x \in M$ is an **accumulation point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A different from x .

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A point $x \in M$ is an **accumulation point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A different from x .

The set $\bar{A}$ of all adherent points of A is called the **closure** of A .

Challenge!

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Which sets are closed?

Challenge!

Problem

Prove the following (Apostol Theorem 3.36). If (M, d) is a metric space, $U \subseteq M$ is open, and $C \subseteq M$ is closed, then $U \setminus C$ is open and $C \setminus U$ is closed.

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Theorem (Apostol Theorem 3.37)

Let (M, d) be a metric space and $A \subseteq S$. Then the following are equivalent:

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- (a) A is closed*
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- (c) A contains all of its accumulation points*

Characterizing closed sets

Theorem (Apostol Theorem 3.37)

Let (M, d) be a metric space and $A \subseteq S$. Then the following are equivalent:

- (a) *A is closed*
- (b) *A contains all of its adherent points*
- (c) *A contains all of its accumulation points*
- (d) *$A = \bar{A}$*

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